LAB PROGRAM-3

```
    abstract class EnergySource {
    int id;
    String name;
    double energy;

    EnergySource(int id, String name, double energy) {
        this.id = id;
        this.name = name;
        this.energy = energy;
    }

    abstract double calculateEfficiency();

    void display() {
        System.out.println("Source ID: " + id);
        System.out.println("Source Name: " + name);
        System.out.println("Energy: " + energy);
        System.out.println("Efficiency: " + calculateEfficiency() + "%");
    }
}

class SolarEnergy extends EnergySource {
    SolarEnergy(int id, String name, double energy) {
        super(id, name, energy);
    }

    double calculateEfficiency() {
        return (energy / 5000) * 100;
    }
}

class WindEnergy extends EnergySource {
    WindEnergy(int id, String name, double energy) {
        super(id, name, energy);
    }

    double calculateEfficiency() {
        return (energy / 8000) * 100;
    }
}

public class Main_1 {
    public static void main(String[] args) {
```

```
        EnergySource e;

        e = new SolarEnergy(101, "Solar", 4000);
        e.display();

        e = new WindEnergy(102, "Wind", 6000);
        e.display();
    }
}
```

```
 ----jGRASP exec: java Main_1
Source ID: 101
Source Name: Solar
Energy: 4000.0
Efficiency: 80.0%
Source ID: 102
Source Name: Wind
Energy: 6000.0
Efficiency: 75.0%

 ----jGRASP: Operation complete.
```

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